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Subject: Information Assurance and Security 2 (IAS2)
Date: June 3, 2026

Protecting Information and Improving Data Security

Today, information is one of the most important things we need to protect. People, businesses, schools, and governments all depend on digital data every day. But as technology grows, so do the dangers that come with it. Hacking, phishing, malware, and identity theft are now very common problems that affect many people around the world. As a future IT professional, I believe it is my responsibility to help make the digital world a safer place.

Information security is important because it keeps our data safe from people who should not have access to it. There are three main ideas behind it. First is confidentiality, which means only the right people can see the data. Second is integrity, which means the data stays correct and is not changed without permission. Third is availability, which means people can access the data when they need it. If any of these three are broken, it can cause big problems like losing money, damaging someone's reputation, or even putting lives at risk in places like hospitals or government offices.

There are many common threats to our digital information. Phishing is when someone sends a fake email or creates a fake website to trick people into giving out their passwords or personal details. Malware and ransomware are harmful programs that can lock your files and ask for money to get them back. A well known example is the WannaCry attack in 2017 that affected more than 200,000 computers in 150 countries. Another threat is social engineering, where attackers trick people instead of hacking systems directly. Weak passwords are also a big problem because they make it easy for attackers to get into accounts. I also personally saw how many people fell for the fake "Free 51GB Data" scam on Labor Day this year, which shows that a lot of us are still not aware enough about these kinds of tricks.

To protect our information, we need to do several things. Using strong and unique passwords for every account is a good start. Turning on multi-factor authentication adds another layer of protection. Encrypting data means that even if someone gets it, they cannot read it. Keeping software updated is also important because old software often has security holes. Organizations should also train their employees regularly so they know how to spot phishing emails and other threats. Having a backup of important data and a plan for when something goes wrong is also very helpful.

As IT professionals, we have a big responsibility. We need to keep learning because new threats come out all the time. We should build security into our systems from the very beginning and not just add it later. We must also be honest and ethical. If we find a problem in a system, we should report it properly and not use it for personal gain. When a security problem happens, we should be open about it and inform the people who are affected right away.

In my opinion, cybersecurity is something that everyone should care about, not just IT people. We all use the internet every day, so we should all know the basics of staying safe online. Better education about cybersecurity, starting from school, can go a long way in preventing attacks. I hope that as I continue my studies and career in IT, I can contribute to making information safer for everyone.